UNIVERSITY OF MORATUWA
Faculty of Engineering
Department of Electronic and Telecommunication Engineering
B.Sc. Engineering
Semester 5 (22 Batch) Examination

# BM3122 MEDICAL IMAGING

Time allowed: *Two* hours November 2025

## ADDITIONAL MATERIAL:

- An information sheet on common constants and physical properties is attached.

## INSTRUCTIONS TO CANDIDATES:

- This paper contains 4 questions on 6 pages.
- This examination accounts for 50% of the module assessment. The total maximum mark attainable is 100. The marks assigned for each question & sections thereof are indicated in square brackets.
- This is a closed-book examination.
- Derivations are not required if they are not explicitly requested for in the question.
- Assume reasonable values for any data not given in or with the examination paper. Clearly state such assumptions.
- If you have any doubt as to the interpretation of the wording of a question, make your own decision, and clearly state it.
- All questions carry equal marks.
- Answer **ALL** questions.

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# Information Sheet

Avogadro's number $= 6.02 \times 10^{23}$
Boltzmann's constant: $k = 1.38 \times 10^{-23}$ J/K
Density of water: $\rho_{\text{water}} = 1,000$ kg/m$^3$
Gyro-magnetic ratio of protons: $\gamma = 267.52 \times 10^6$ rad s$^{-1}$T$^{-1}$
Plank's constant: $h = 6.63 \times 10^{-34}$ Js

**Some basic relationships** – *Symbols have their usual meanings*

De Broglie's relationship: $\Delta E = hf$

For spins in a strong magnetic field: $N_{\text{parallel}} - N_{\text{anti-parallel}} = N_{\text{total}} \dfrac{\gamma h B_0}{4\pi k T}$

Net magnetization for spins in a strong magnetic field: $\mathbf{M}_0 = \dfrac{\gamma h}{4\pi}\left(N_{\text{parallel}} - N_{\text{anti-parallel}}\right)$

Image intensity of MRI pixel acquired using a Gradient Echo sequence: $I(x,y) \propto \rho(x,y) \dfrac{\left(1 - e^{-\text{TR}/\text{T}_1}\right)\sin\alpha}{1 - e^{-\text{TR}/\text{T}_1}\cos\alpha} e^{-\frac{\text{TE}}{\text{T}_2^*}}$

Optimized tip angle for a given TR: $\alpha_{\text{Ernst}} = \cos^{-1} e^{-\frac{\text{TR}}{\text{T}_1}}$

The intensity of an axial MRI slice acquired using Spin Echo sequences: $I(x,y) \propto \rho(x,y)\left(1 - e^{-\frac{\text{TR}}{\text{T}_1}}\right) e^{-\frac{\text{TE}}{\text{T}_2}}$

Radioactivity of a radioactive isotope: $Q = \dfrac{dN}{dt} = \lambda N$

The number of radioactive nuclei at time, $t$: $N(t) = N(t=0)e^{-\lambda t}$

The effective radiotracer half-life: $\tau_{1/2,\text{eff}} = \dfrac{\tau_{1/2} \cdot \tau_{1/2,\text{bio}}}{\tau_{1/2} + \tau_{1/2,\text{bio}}}$

Gaussian line spread function: $LSF(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} exp\left(-\dfrac{(x - x_0)^2}{2\sigma^2}\right)$

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**Question 1.**

(a) The point spread function (PSF) of an imaging device is the response of the system to an ideal point target and it is a measure of spatial resolution.

i. What is the spatial resolution of an imaging system? [1]

ii. For the one-dimensional objects $O(x)$ and PSFs $h(x)$ shown in Figure Q1, draw the resulting projections $I(x)$. [6]

iii. State which object in Figure Q1 is affected most by the action of h(x) with reasoning. [3]

(b) A small amount of a radioactive material called a radiotracer is injected to the patient in nuclear medicine imaging.

i. State **two** advantages and **two** disadvantages of nuclear medicine imaging. [4]

ii. Briefly explain the annihilation coincidence detection process in PET. [3]

iii. State **two** advantages of a hybrid PET/CT scanner. [2]

iv. Consider an experiment that uses dual radionuclides, technetium-99m and thallium-201. At $t = 0$, we have equal number of atoms of both radionuclides. When will the count rate of technetium-99m be smaller than 20 percent of the total count rate? Half-life of technetium-99m and thallium-201 are 6 h and 73 h respectively. [6]

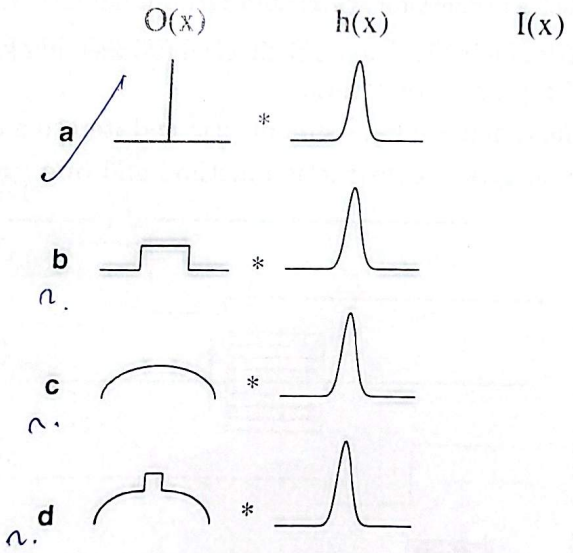



Figure Q1: One-dimensional objects O(x) and PSFs h(x)

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## Question 2.

(a) A radiography image is a transmission image showing the amount of X-ray attenuation along a path through the body.

i. Briefly explain the functions of main components of a planar X-ray radiography system. [6]

ii. State **two** methods to reduce the size of the effective focal spot of an X-ray system. [2]

iii. What are the effects of a smaller effective focal spot on the image. [3]

iv. X-rays with intensity $I_0$ are incident upon a 1 cm thick material for which $\mu = 0.3$ cm$^{-1}$. What intensity will be detected? [3]

(b) X-ray Computed Tomography (CT) produces tomograms of the distribution of linear attenuation coefficient, expressed in Hounsfield units.

i. What is the importance of defining CT numbers (Hounsfield Units) in CT imaging? [2]

ii. A calibration procedure for a CT scanner measures the Hounsfield number for air as -1100 and for water as 10. A linear calibration curve for this scanner is defined as,

$$ah_m + b = h$$

where $h_m$ is the measured CT number and $h$ is the actual value. Calculate the constants $a$ and $b$ to calibrate this scanner. You may use the standard values of CT numbers for water and air. [4]

(c) Image reconstruction in CT is a mathematical process that generates tomographic images from X-ray projection data.

i. What is the difference between analytic and iterative reconstruction? [3]

ii. State **one** analytic reconstruction method and **one** iterative reconstruction method. [2]

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## Question 3.

(a) Magnetic Resonance Imaging (MRI) uses differences in magnetic properties of atomic nuclei to generate a spatial map of hydrogen nuclei in different tissues.

i. Give **three** factors that can decide the intensity of an MRI image. [3]

ii. We require to have a slice thickness of 3 mm in an MRI system where the z-gradient strength $G_z = 0.3$ mT/cm. What is the RF range that should be excited? [4]

iii. State **two** methods to reduce slice thickness. [2]

(b) Figure Q3 shows a basic gradient echo pulse sequence used for MRI image acquisition.

i. Explain briefly the timing and the purpose of each gradient line of the pulse sequence. [3]

ii. Show how this pulse sequence can be compressed to save time without affecting the image. [3]

iii. An MRI image slice of matrix size 256 × 256 is acquired with parameters, repetition time (TR) = 4500 ms and echo time (TE) = 18 ms. Calculate the time required for acquisition of the whole slice. [2]

iv. Show how the pulse sequence in the Figure Q3 can be modified to acquire multiple slices within the calculated time in (b) iii. [4]

v. What is the main advantage of spin echo over gradient echo sequence? [2]

vi. What is the mechanism to achieve the advange mentioned in (b) v.? [2]

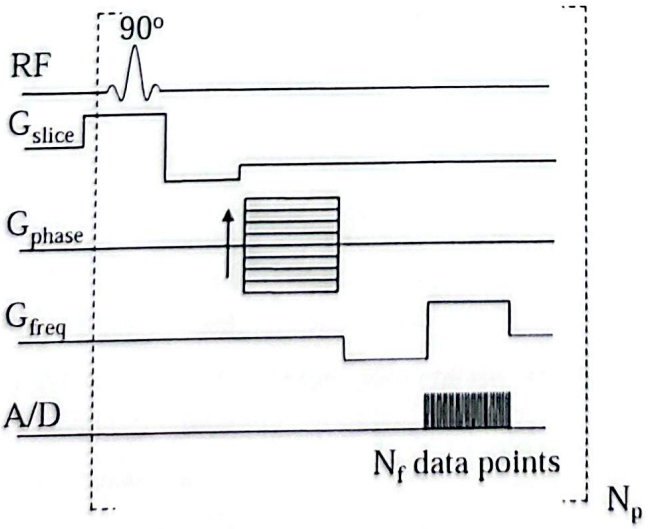



Figure Q3: Gradient Echo Pulse Sequence

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**Question 4.**

(a) Medical ultrasound is the second most commonly used imaging modality in clinical imaging.

i. What is the physical reason to avoid an air gap between the transducer and the patient? How can it be avoided? [3]

ii. Briefly explain speckles in ultrasound imaging. [3]

iii. Explain the trade-off between resolution and penetration in ultrasound imaging and state **two** mechanisms for addressing this trade-off. [4]

iv. Consider a phased array ultrasound system operating at 2 MHz and capable of B-mode imaging. Calculate the depth of penetration that can be acheived with an 80dB dynamic range. Assume an attenuation coefficient of 1.0 dB $cm^{-1}$ $MHz^{-1}$. [4]

v. In (a) iv., what is the maximum frame rate acheivable, for a $256 \times 256$ B-mode image? Assume a speed of sound $c = 1480$ m/s. [3]

(b) Optical Coherence Tomography (OCT) is a non-invasive imaging technique very similar to clinical ultrasound.

i. What is the property represented by a pixel in OCT image? [2]

ii. Give **one** main advantage and **one** main disadvantage of OCT compared to ultrasound imaging. [3]

iii. Briefly explain **one** major application of OCT. [3]

End of Paper.



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